

Denis Treskunov

Software Engineer · Seattle, WA

Problem solver, lifelong learner, innovator.

Looking for new opportunities to create impact and make a difference.



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Education

01/2004 – 06/2008

Bachelor of Science, Mathematics

University of California · Berkeley, CA

Computer Science Minor

Experience

04/2020 – PRESENT

Software Engineer, Principal

Salesforce · Seattle, WA

- Implemented support for **Multi-factor Authentication (MFA)** for signing in to Tableau Online.
- Developed **EasySSL**, a **library for zero-downtime updates** of server and client TLS certificates.
- Led a major, **6-month refactoring project** to improve reliability of the sign-in service to and reduce technical debt. The refactoring reduced the LoC (Lines of code), as well as Cognitive and Cyclomatic complexity, while resolving long-standing defects.
- Created a middleware service to allow Tableau products to use Salesforce's **LLM / GenAI** service.

10/2011 – 04/2020

Software Engineer, Staff

Tableau Software · Seattle, WA

- Championed the Infrastructure as code (IaC) concept. Created a **versioned, idempotent deployment service** to deploy Tableau's SaaS products and their dependencies. Technologies used: **Puppet** to manage dependencies and roll out services, and **Rundeck** to allow the Ops team to execute pre-defined runbooks from a web app.
- Created the **login page** for Tableau Online handling **~800k daily users at 99.99% availability**. This service supports authentication via **SAML**, **OIDC**, and **Mutual TLS**, and is distributed globally across multiple regions for performance and resiliency. Technologies used: **Spring Boot**, **Cassandra**, **DynamoDB**.
- Migrated on-premise infrastructure into AWS. Created build pipelines for triggering deployments after code merges. Technologies used: **Terraform**, **GitLab CI/CD**.
- Architected a live migration of the user database from an on-premise Cassandra cluster into DynamoDB. To validate the new schema, Java instrumentation was used to detect any differences between the two databases.
- Refactored and upgraded SAML logic into a separate library, **fixing 16k compile errors**.

12/2008 – 10/2011

Software Development Engineer

Amazon · Seattle, WA

- Implemented a **bytecode-level record/replay framework** that was used by the team for regression testing against captured production data. The framework enabled **nearly 100% code coverage for 22k lines of business logic**.
- Rewrote a Perl/Mason UI in **Google Web Toolkit** to resolve reliability, security and performance concerns. Eliminated page load fatals and **reduced latency and bandwidth by 15x**.
- Participated in the design and implementation of a next-generation **distributed workflow engine that was 3x more efficient** than the engine it replaced, horizontally scalable, and multitenant.
- Was involved in a multi-year effort to **rearchitect and migrate 126k lines** of C++ business logic to Java and to remove other legacy dependencies.

Open Source

- **EasySSL**: Public Java library allowing zero-downtime updates of server and client TLS certificates. Used internally at Tableau.
- Tableau importers for web data sources:
 - **cljs-wdc**: ClojureScript wrapper for the Tableau Web Data Connector library.
 - **wta-wdc**: data from **Washington Trail Association**.
 - **dhis2-wdc**: data from **DHIS2**, an international Health management software platform.
- **vosk-api**: adaptation of the Kaldi speech recognition library to WebAssembly. **Demo here**.

Skills

- Distributed systems
- Technical leadership
- Java (expert), TypeScript, Terraform (expert), AWS (Elastic compute, DynamoDB, CloudWatch)
- Infrastructure as code (IaC)